The Stanley Controller and Rear-Wheel Feedback

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Where this fits:

- **Builds on:** Lateral and heading error; Path coordinates (Frenet frame).
- **Leads to:** Comparing and choosing trackers.
- **Compare with:** Pure pursuit.

1. Overview

Key point: Pure pursuit steers for a point ahead and never looks at the path where the car is, so it turns early at the corners and its look-ahead must be retuned for speed. The Stanley controller steers the front wheels from the errors at the front axle, and rear-wheel feedback turns the car from the errors and the path's curvature at the rear axle; both follow the 10 m arc with no look-ahead to tune.

Pure pursuit (G-3150) brings the car from 1.0 m off the course to within 0.1 m after 7.4 m. What still breaks is the corners. Pure pursuit steers for the look-ahead point (G-3152), the course point 4 m ahead, so it starts to turn before the arc does and straightens before the arc ends: its rear axle runs 0.12 m inside the arc and 0.17 m wide after it. And the right look-ahead distance (G-3151) changes with speed.

Figure 1 replays pure pursuit on the course. Watch its lateral error at the arc's two ends. The last frame adds two trackers that measure the errors at the car's own axles; neither shows a bump at the corners.

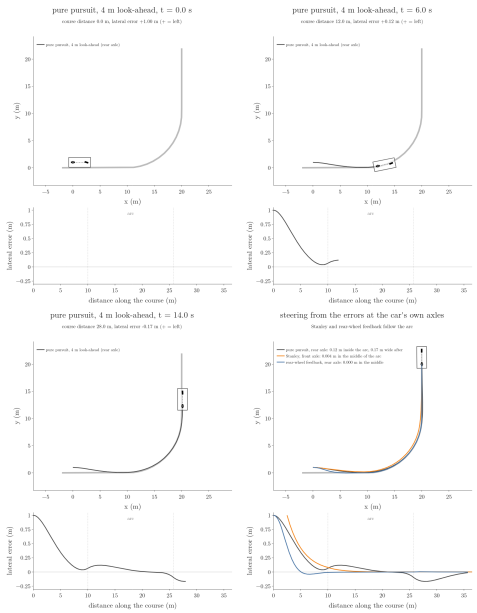


Figure 1: Pure pursuit with a 4 m look-ahead at 2 m/s (grey) on the course, one frame per 0.5 s. Top: the paths. Bottom: the lateral error along the course (+ = left), with the arc between the dotted lines; pure pursuit's rear axle runs 0.12 m inside the arc and 0.17 m wide after it. The last frame adds Stanley (orange; error measured at its front axle) and rear-wheel feedback (blue; at its rear axle): 0.004 m and 0.000 m in the middle of the arc

The car and its course are RO-071's:

- a car with a wheelbase (G-3127) of 2.5 m that steers its front wheels by at most 26.6°, moved by the kinematic bicycle model (G-3128), with no tyre slip;
- the course: 10 m straight along x from (0, 0), a 90° left arc of radius 10 m, then 10 m straight;
- the start: rear axle at (0, 1.0), 1.0 m left of the line, heading along it;
- 2 m/s, and a new steering command 20 times a second.

Every run in this Note is our own Python model of the two laws on this car (the notebook `RO-072-stanley-and-rear-wheel-feedback.ipynb`, with RO-071's simulator), not the code of a real car.

The lateral error (G-3135), also called the cross-track error, is the signed distance from the course, positive to the left, and the heading error (G-3136) is the heading minus the course's direction at the nearest point. Both depend on which point of the car we measure from: pure pursuit and rear-wheel feedback use the rear axle, Stanley the front axle.

This Note builds the two trackers:

- the Stanley controller: a heading term and a cross-track term, both at the front axle (Section 2);
- why its front-axle error shrinks by the same fraction every second, and what that means at 2 and 5 m/s and on the arc (Section 3);
- tuning Stanley: its gain and its softening speed (Section 4);
- rear-wheel feedback, which also steers the car backwards (Section 5);
- where the two run (Section 6).

2. Stanley: steer the front wheels from the front axle

> **Key point:** Stanley measures the lateral error and the heading error at the front axle, and turns the front wheels parallel to the course (the heading term), then further towards it (the cross-track term), so that they aim at a course point a set distance ahead, the speed plus a softening speed over a gain: at the start, 3 m ahead, so the wheels turn 18.4° to the right at 2 m/s.

2.1 Why the front axle

Pure pursuit's corner errors come from aiming at a point ahead. The front wheel is the wheel that steers, and it rolls the way it points. Point it at the course and the front axle heads for the course at once; point it along the course and the front axle runs along it. A cyclist riding along a painted line does the same: the front wheel is kept on the line, and the rest of the bicycle follows.

So Stanley measures its errors where the steering acts: the lateral error is "the lateral distance of the center of the vehicle's front wheels from the nearest point on the trajectory" (Thrun et al. §9.2; Hoffmann et al. §III.A). At the start the front axle is 2.5 m ahead of the rear axle,

at (2.5, 1.0), so its nearest course point is (2.5, 0): its lateral error is 1.0 m, and its heading error is 0, because the car points along the course.

2.2 The heading term and the cross-track term

Stanley turns the wheels in two steps (Snider §2.3; Thrun et al. §9.2):

1. **The heading term** turns the wheels from the car's heading to the course's direction at the front axle's nearest point, so that with no lateral error the wheels point along the course; the term is minus the heading error.
2. **The cross-track term** turns them further towards the course, more the farther away the front axle is, by the angle whose tangent is the gain times the lateral error over the speed plus a softening speed.

The cross-track angle has a simple picture. A wheel turned by that angle from the course's direction heads for the course point a fixed distance ahead of the nearest point, because the tangent of the angle is the lateral error over that distance. With:

- e, the front axle's lateral error, positive to the left: 1.0 m at the start;
- θ_e, its heading error, positive to the left: 0 at the start;
- v, the car's speed, 2 m/s;
- k, the gain, 1.0 per s;
- k_s, the softening speed, 1.0 m/s (Section 4.2);
- d, how far ahead of the nearest point the wheels aim, in m;
- δ, the steering angle, positive to the left, at most 26.6° either way.

$$d = (1 + 2)/1 = 3 \text{ m}$$

$$\tan(\text{cross-track term}) = 1.0/3$$

$$d = \frac{k_s + v}{k}$$

$$\tan(\text{cross-track term}) = \frac{e}{d} = \frac{k\,e}{k_s + v}$$

The law adds the two terms, with signs that steer towards the course (Hoffmann et al. eqs. 5 and 9; Snider eq. 5; Paden eq. V.15):

$$\delta = -\theta_e - \tan^{-1}\frac{k\,e}{k_s + v}$$

and the result is held within ±26.6°, the car's limit. At the start, at 2 m/s:

$$\delta = -0 - \tan^{-1}(1 \times 1.0/3)$$

$$= -\tan^{-1}(0.333) = -18.4^\circ$$

The wheels turn 18.4° to the right, towards the course. At 5 m/s the aim point is 6 m ahead and the wheels turn 9.5°. Hoffmann et al. write the same law with the course's direction seen from the car and the lateral error positive to the right; with our signs, positive to the left, it is the line above. The steering limit is part of the law: the steering angle is held at the limit whenever the formula asks for more (Hoffmann et al. eq. 5).

The rule is the **Stanley controller** (G-3160), also called front-wheel feedback (Paden §V.A.3): the steering law of Stanford University's car Stanley (Thrun et al. 2006; Hoffmann et al. 2007).

Figure 2 shows both terms half a second into the 2 m/s run. The car now points 5.4° away from the course, to the right, and its front axle is 0.71 m off: the heading term turns the wheel 5.4° back, parallel to the course (purple), and the cross-track term turns it a further 13.3° towards the aim point 3 m ahead (orange).

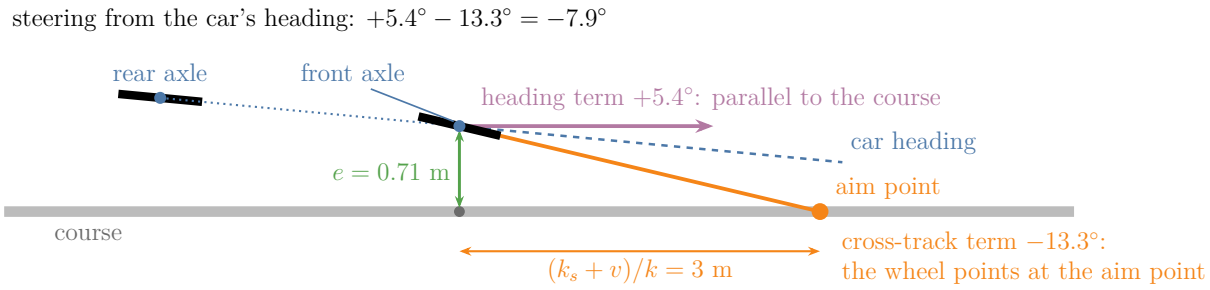


Figure 2: The Stanley law 0.5 s into the 2 m/s run. The front axle is 0.71 m left of the course (green), and the car heads 5.4° to the right of the course's direction. The heading term turns the front wheel +5.4°, parallel to the course (purple); the cross-track term turns it a further −13.3°, so that it points at the course point 3 m ahead of the nearest point (orange). Steering from the car's heading: +5.4° − 13.3° = −7.9°

The law says where to point the wheels at each tick. Whether the front axle then comes back without swinging, as pure pursuit's rear axle did, is the next question.

3. Why the front-axle error shrinks by the same fraction every second

> **Key point:** With the wheels turned by Stanley's two terms, the front axle moves at the cross-track angle to the course, so its lateral error, its distance from the course, shrinks in proportion to itself: by the same fraction every second, set by the gain k, almost whatever the speed, and on the chapter's car, whose wheels turn at once, it never swings

past the line. At 2 m/s the error halves in 1.03 s; at 5 m/s in 0.81 s, though the car drives twice as far.

3.1 The front axle heads for the course

At the start the front wheel points 18.4° to the right of the course, so the front axle moves towards the course at 2 m/s times the sine of 18.4°. The heading term keeps cancelling the car's own turning, so at every tick the front wheel makes the cross-track angle with the course, and only that angle decides how fast the error shrinks (Hoffmann et al. eq. 7; Thrun et al. eq. 8; Paden eq. V.13). With the symbols of Section 2.2, the lateral error e, the gain k, the softening speed k_s and the speed v, and:

- $\dot{e}$, how fast the front axle's lateral error changes, in m/s;
- v, the speed, also stands for the front wheel's speed, which is close (exactly, the rear axle's speed divided by the cosine of the steering angle).

At the start, at 2 m/s:

$$\dot{e} = -2 \times \sin(18.4^\circ)$$

$$= -2 \times 0.316 = -0.632 \text{ m/s}$$

$$\dot{e} = -v \sin\left(\tan^{-1} \frac{k\,e}{k_s + v} \right)$$

For a small error the sine of a small angle is the angle, and the angle is about its tangent:

$$\dot{e} \approx -\frac{k\,v}{k_s + v}\, e$$

The error shrinks at a rate in proportion to itself, which makes it a first-order system (G-2987): it closes the same fraction of its gap in every second and can never overshoot. Without the softening speed, the rate is the gain k itself, 1 per s, whatever the speed, so after a time t in seconds the error from a 1.0 m start is (Thrun et al. eq. 9; Hoffmann et al. §III.A):

$$e(t) = 1.0 \times \exp(-1 \times t)$$

$$e(t) = e(0)\, \exp(-k\,t)$$

With the softening speed of 1 m/s the rate is a little lower:

$$\text{rate at 2 m/s} = 1 \times 2/3 = 0.667 \text{ per s}$$

$$\text{rate at 5 m/s} = 1 \times 5/6 = 0.833 \text{ per s}$$

The time constant (G-2988), one over the rate, is 1.5 s at 2 m/s and 1.2 s at 5 m/s. The time to halve the error is the natural log of 2, 0.693, over the rate:

$$0.693/0.667 = 1.04 \text{ s at 2 m/s}$$

$$0.693/0.833 = 0.83 \text{ s at 5 m/s}$$

When the error is large the arctangent nears 90° and is held at the steering limit, so the car drives straight at the course and the error shrinks no faster than the speed allows (Hoffmann et al. §III.A).

3.2 At 2 and 5 m/s

Figure 3 runs Stanley at both speeds for 3 s. Watch the orange arrow: at the start it points well right of the purple one, and it swings back towards it as the front axle closes in. Below, the solid error curves lie on the dashed predictions of Section 3.1 and never cross zero.

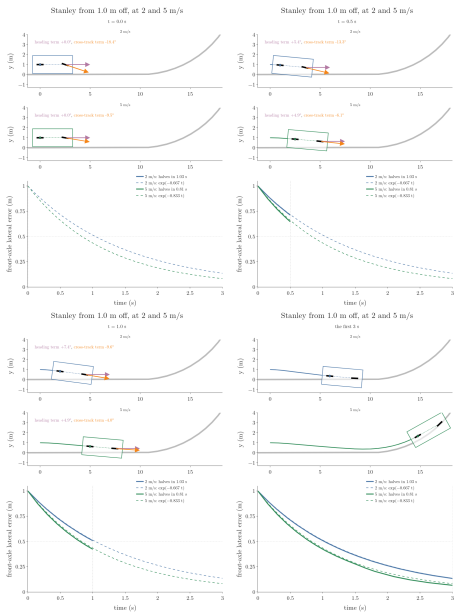


Figure 3: Stanley from 1.0 m left of the course at 2 m/s (top) and 5 m/s (middle), one frame per 0.1 s for 3 s. At each front axle: purple, the course's direction at the nearest point, where the heading term alone would point the wheel; orange, the wheel's commanded direction, turned further towards the course by the cross-track term; the angles are written in each panel. Bottom: the front axle's lateral error against time, with the predictions $\exp(-0.667\ t)$ and $\exp(-0.833\ t)$ dashed. The error halves in 1.03 s at 2 m/s and 0.81 s at 5 m/s

The runs (notebook):

Speed	Error halves after	Front axle driven meanwhile	Within 0.1 m after	Past the line
2 m/s	1.03 s	2.0 m	3.44 s	none
5 m/s	0.81 s	4.0 m	2.55 s	none

The halving times are the predictions of Section 3.1 to within 0.02 s. At 5 m/s the arc starts before the error is below 0.1 m; the arc bends towards the car, which is on its inside, and shortens the last part of the return (2.55 s against 2.76 s predicted for a straight course). Stanley's error shrinks in a similar time at both speeds (1.03 s and 0.81 s to halve), so the fast car needs about twice the road; pure pursuit's path did not depend on the speed at all (pure pursuit's speed check).

3.3 On the arc: the front axle stays on it, the rear axle cuts inside

On the 10 m arc the front axle's lateral error falls to 0.004 m by the middle of the arc (notebook). There the cross-track term is about zero, so the heading term alone does the steering. Paden et al. (§V.A.3) point out that this term is then not a heading error to be removed: on a curve the front wheel must point along the course, while the car's body, with its rear axle on a smaller circle, points 14.5° to the outside of the course's direction at the front axle (worked out below; Figure 4), and the heading term holds exactly the steering angle that the curve needs.

The rear axle then runs on a smaller circle. The front axle is on the course, 10 m from the arc's centre, and the line from the rear axle to the centre is square to the car's body, which is 2.5 m long (Figure 4). With:

- R_f, the front axle's distance from the arc's centre, 10 m;
- L, the wheelbase, 2.5 m;
- R_r, the rear axle's distance from the centre, in m;
- δ, the steering angle on the arc.

$$R_r = \sqrt{10^2 - 2.5^2}$$

$$= 9.682 \text{ m}$$

$$\sin\delta = 2.5/10 = 0.25$$

$$\delta = 14.5^\circ$$

$$R_r = \sqrt{R_f^2 - L^2}$$

$$\sin\delta = L/R_f$$

So the rear axle runs inside the arc by:

$$10 - 9.682 = 0.318 \text{ m}$$

which the simulation confirms: 0.317 m in the middle of the arc (notebook).

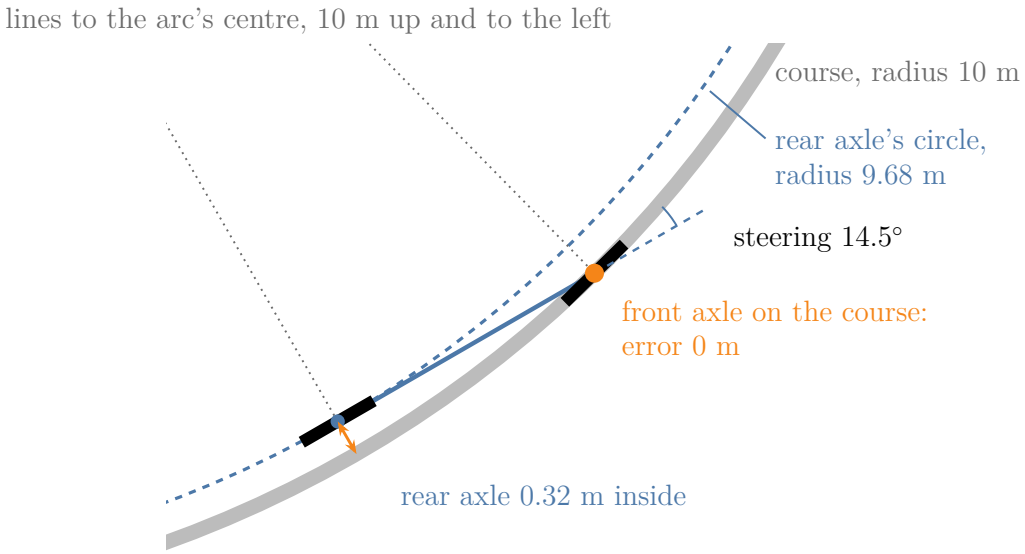


Figure 4: Stanley on the 10 m arc once settled. The front axle (orange) stays on the course, 10 m from the arc's centre; the rear axle, 2.5 m behind it, runs on a circle of radius 9.68 m, 0.32 m inside the course; the steering is 14.5°. Only the part of the arc near the car is drawn; the dotted lines run to the centre, 10 m away

The gotcha: an error depends on the point of the car it is measured at. Stanley's front axle is on the arc while its rear axle is 0.32 m inside, so two trackers can be compared only at the same point of the car, which is why the chapter's comparison measures them all at the rear axle. Stanley's one setting so far was the gain $k = 1$ per s. The next section asks how to choose it.

4. Tuning Stanley: the gain and the softening speed

> **Key point:** A larger gain k brings the front axle back sooner, until the steering hits its limit; once we give the car RO-071's steering lag of 0.4 s, the delay of real steering, at 5 m/s gains of 2.5 and 5 per s swing 0.24 and 0.46 m past the line, so the gain has an upper limit. The softening speed k_s, 1 m/s, keeps the gain on the lateral error, the front axle's distance from the course, below 1 per m when the car is slow, so noise in the measured error does not shake the wheels.

4.1 The gain

The gain sets the rate at which the front axle returns. On the chapter's car, whose wheels turn at once, more gain only means a faster return: at 5 m/s, with k of 0.5, 1, 2.5 and 5 per s, the front axle is within 0.1 m after 32.3, 15.4, 7.8 and 5.1 m along the course, and at 5 per s the wheels sit at full lock at the start (notebook). Section 3.1 says why: the error shrinks in proportion to the gain, and nothing in that kinematic model can swing.

A real car's steering does lag, and Snider, testing on a full car simulator with delays added to the steering (Snider §1), found an upper limit on the gain (§2.3.1). So, as RO-071 did, we change the chapter's car for this sweep: it gets RO-071's steering motor, which turns the wheels with a time constant of 0.4 s. Without the lag, no gain would show the limit. Figure 5 sweeps the gain again at 5 m/s. Watch the red and orange curves below: they come back fastest, but cross the line and swing back; on the arc all four swing.

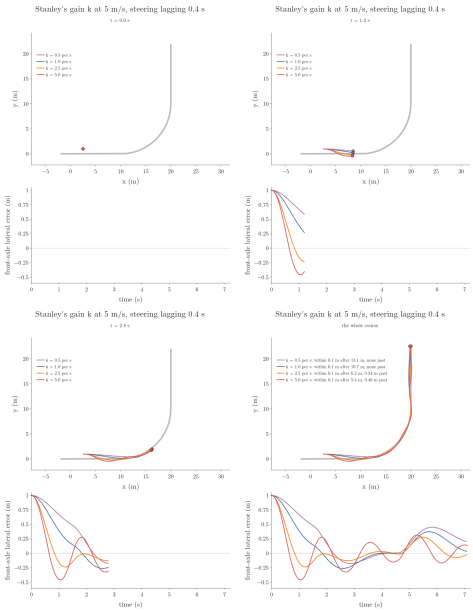


Figure 5: Stanley's gain k swept at 5 m/s with steering that lags 0.4 s (added to the chapter's car for this sweep), from 1.0 m left of the course; one frame per 0.2 s: k = 0.5 (purple), 1 (blue), 2.5 (orange) and 5 per s (red). Top: the paths of the front axles. Bottom: the front axle's lateral error against time. Last frame: on the first straight k = 0.5 and 1 do not cross the line, k = 2.5 swings 0.24 m past it and k = 5 swings 0.46 m past; on the arc every gain swings

Gain k	Within 0.1 m after (front axle)	Past the line, first straight	Largest on the arc	Largest after the arc
0.5 per s	13.1 m	none	0.48 m, inside	0.45 m
1 per s	10.7 m	none	0.26 m, outside	0.37 m
2.5 per s	6.2 m	0.24 m	0.13 m, outside	0.28 m
5 per s	5.4 m	0.46 m	0.32 m, outside	0.31 m

Snider (§2.3.1) puts it as "An upper limit exists to guarantee stability." The harder the law corrects, the further the late wheels overshoot. On the arc every gain now leaves 0.13 to 0.48 m, where on the chapter's car gains of 2.5 and 5 per s left 0.04 m or less (notebook). Stanford's car met both problems with extra terms (Section 4.3).

4.2 The softening speed

The cross-track term divides by the speed. Slowly, the gain on the lateral error becomes huge: at 0.5 m/s, with $k = 1$ per s and no softening, it is 2 per m, and at a standstill it has no limit. Measured positions jitter, so the measured lateral error jitters too, and a large gain passes the jitter to the wheels (Hoffmann et al. §III.A). A jump of 0.05 m in the measured error at 0.5 m/s turns the wheels by (notebook):

$$\tan^{-1}(1 \times 0.05/0.5) = 5.7°$$

Adding a **softening speed** (G-3161) k_s to the speed in the denominator caps the gain at the gain k over k_s, 1 per m, "permitting control to be arbitrarily soft at low speeds"; Stanford's car used 1 m/s (Hoffmann et al. §III.A). With it, the same jump at 0.5 m/s turns the wheels by:

$$\tan^{-1}(1 \times 0.05/1.5) = 1.9°$$

Figure 6 draws the gain on the lateral error against speed, with and without the softening speed.

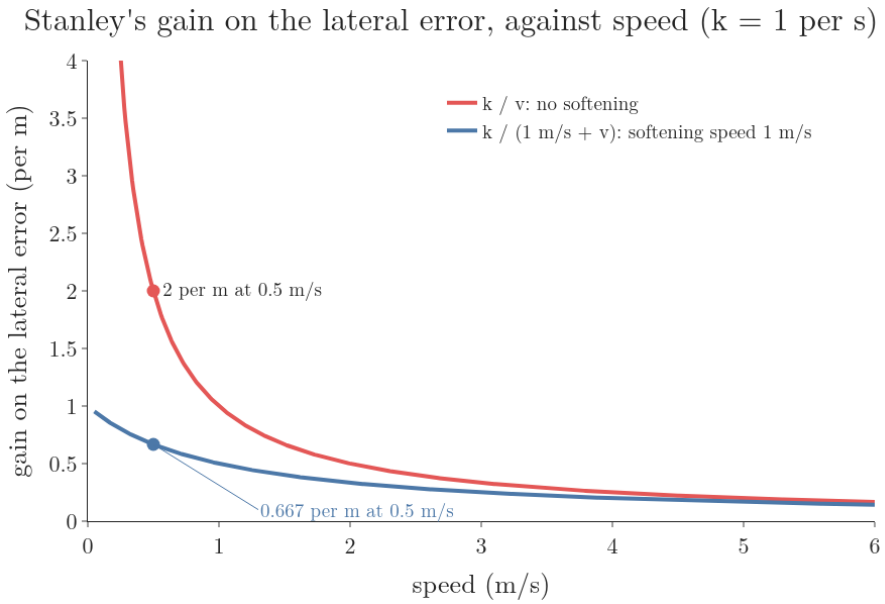


Figure 6: Stanley's gain on the lateral error against speed, for k = 1 per s. Red: k / v, without a softening speed, which grows without limit as the car slows (2 per m at 0.5 m/s). Blue: k / (1 m/s + v), with a softening speed of 1 m/s, which never exceeds 1 per m (0.667 per m at 0.5 m/s)

The cost shows at higher speeds: at 2 m/s the softened gain is 1/3 per m instead of 1/2, so the aim point is 3 m ahead instead of 2 m and the error shrinks at 0.667 per s instead of 1 per s (Section 3.1). At 5 m/s the two differ by less than a fifth.

4.3 What Stanford's car added

> **Extra:** On the real car, tyres slip and the steering column, the steering motor and the computer all add delay. Stanford's final law (Hoffmann et al. eq. 9; Thrun et al. §9.2) adds three terms to the two of Section 2:
>
> - a damping term on the yaw rate, the rate at which the car's heading turns, because

the tyres damp the car's yawing less as the speed grows;
- a damping term on how fast the steering angle changes, against the motor's delay;
- a steady heading offset on curves, worked out from a bicycle model with tyre slip, so that the car can lean its heading into a turn as a slipping car must.

Hoffmann et al. report that the law ran "without fault on Stanley for hundreds of miles of testing" and in the race. MATLAB's Stanley block offers the same choice: a kinematic bicycle model, "suitable for path following in low-speed environments", and a dynamic one for high speeds, which also takes the path's curvature, the yaw rate and the steering angle, with gains on the last two (MathWorks Lateral Controller Stanley documentation).

Stanley handles the corners and its gain has a clear meaning. But it has one blind spot that matters for a car that parks: driving backwards.

5. Rear-wheel feedback: the rear axle, the path's curvature, and reverse

Key point: Rear-wheel feedback asks for a turn rate with three parts: the turn the path's curvature needs, a part against the heading error, how far the car points off the course's direction, and a part against the rear axle's lateral error, its distance from the course; on the 10 m arc the first part alone holds the car on the line. Near the line it is a spring and a damper over distance, with a damping ratio of 0.707 for our gains, and it works driving backwards, where Stanley turns the car round.

5.1 Why the rear axle: driving backwards

The car of the Hybrid A* planner ended its route by backing into a parking bay. Backwards, Stanley's front axle trails behind, and pointing the front wheels at the course turns the car the wrong way: Paden et al. (§V.A.3) warn that Stanley "is not stable in reverse making it unsuitable for parking". Stanford's car reversed with a variant that uses the rear wheels as the guiding wheels (Hoffmann et al. §III.A): measure at the axle that leads, which is the idea of this section.

Figure 7 tries it. The car faces along the first straight, 1.0 m left of it, and drives backwards at 2 m/s for 8 s. Watch the orange car: it swings round until it faces the other way, 9.0 m to the right of the line. The blue car steers by rear-wheel feedback, built in Sections 5.2 and 5.3: it is within 0.1 m after 2.1 s, having driven 4.1 m back.

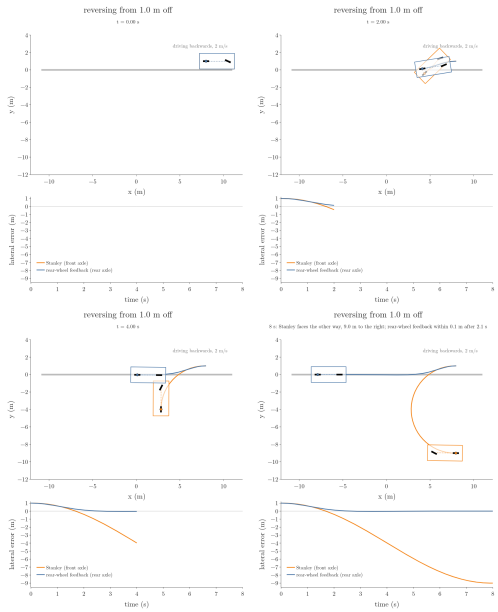


Figure 7: Reversing from 1.0 m left of the first straight at 2 m/s for 8 s; one frame per 0.25 s. Orange: Stanley, steering from its front axle, which now trails; the car swings round until it faces the other way, 9.0 m to the right of the line. Blue: rear-wheel feedback, steering from its rear axle; it returns within 0.1 m after 2.1 s, 0.04 m past the line at most. Bottom: the lateral error of each against time

5.2 Three parts of a turn rate

Rear-wheel feedback (G-3162) measures its errors at the rear axle and asks for a turn rate, the rate at which the car's heading should turn, written ω as for the differential drive (Paden §V.A.2). Its three parts (Paden eq. V.12):

1. **the path-curvature part,** the turn rate that keeps a car on the line turning with the course: the speed times the path curvature (G-3129), corrected for the car being off the line or askew;
2. **the heading part,** against the heading error, in proportion to the speed;
3. **the lateral part,** against the lateral error, in proportion to the speed.

With:

- e, the rear axle's lateral error, positive to the left: 1.0 m at the start;
- θ_e, the heading error at the rear axle's nearest point, in rad: 0 at the start;
- κ, the course's curvature at that point: 0 on the straights, 0.1 per m on the arc;
- v, the speed, 2 m/s (negative when driving backwards);
- k_θ, the heading gain, 1.0 per m;
- k_e, the lateral gain, 0.5 per m^2;
- ω_κ, ω_θ and ω_e, the path-curvature, heading and lateral parts, in rad/s;
- ω, the turn rate asked for, their sum, in rad/s;
- L, the wheelbase, 2.5 m, and δ, the steering angle.

$$\omega_\kappa = \frac{v\kappa\cos\theta_e}{1-\kappa e}$$

$$\omega_\theta = -k_\theta|v|\,\theta_e$$

$$\omega_e = -k_e\,v\,\frac{\sin\theta_e}{\theta_e}\,e$$

$$\omega = \omega_\kappa + \omega_\theta + \omega_e$$

The bicycle model turns the turn rate into a steering angle: the car turns at its speed times the curvature it drives, and that curvature is the tangent of the steering angle over the wheelbase (pure pursuit's steering step):

$$\delta = \tan^{-1}\frac{L\,\omega}{v}$$

The units of the gains make each part a turn rate: k_θ times a speed in m/s and an angle in rad gives rad/s when k_θ is in 1/m, and k_e times a speed and a distance in m gives rad/s when k_e is in $1/\text{m}^2$. The values 1.0 and 0.5 are PythonRobotics' defaults (`KTH`, `KE` in `rear_wheel_feedback_control.py`). The fraction $\sin\theta_e/\theta_e$ is close to 1 for small angles (0.993 at 0.2 rad) and is 1 at $\theta_e = 0$. The divisor $1-\kappa e$ is 1 on the line; off the line it corrects for the nearest course point moving faster or slower than the car, as the path coordinates of the errors' motion show (Paden eq. V.10).

At the start, on the straight, the curvature part and the heading part are 0, and the fraction $\sin\theta_e/\theta_e$ is 1:

$$\omega = 0 - 0 - 0.5 \times 2 \times 1 \times 1.0$$

$$= -1.0 \text{ rad/s}$$

$$\delta = \tan^{-1}(2.5 \times (-1.0)/2)$$

$$= \tan^{-1}(-1.25) = -51.3°$$

More than the car can steer, so the wheels go to the limit, −26.6°.

On the arc, on the line, the lateral and heading parts are 0 and the curvature part does all the turning:

$$\omega = 2 \times 0.1 = 0.2 \text{ rad/s}$$

$$\delta = \tan^{-1}(2.5 \times 0.2/2) = 14.0°$$

the steering of the rear axle's 10 m circle. The curvature part is the path's own turn, sent before any error appears: feedforward (G-3070), the same idea as feedforward steering from the path's curvature. The other two parts are feedback. RO-070's demonstration driver, the line rule at the nearest point, turned the car at -0.5 rad/s per metre of lateral error and -2 rad/s per radian of heading error: rear-wheel feedback at 2 m/s without its curvature part, with half its lateral gain. Lacking the curvature part, it had to run up to 0.39 m outside the arc to turn with it (measuring a run).

Figure 8 drives the course with it at 2 m/s and shows the three parts at each frame. Watch the bars: at the start the lateral part (orange) is large and the total is held by the steering limit; on the arc the green curvature part alone carries the turn, and the other two are near zero.

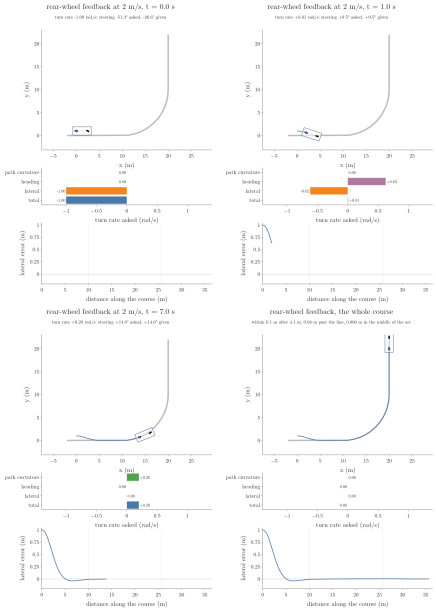


Figure 8: Rear-wheel feedback on the course at 2 m/s from 1.0 m left of the line; one frame per 0.5 s. Top: the car and its rear-axle path. Middle: the turn rate it asks for, as its three parts, path curvature (green), heading (purple) and lateral (orange), and their total (blue), with the steering asked and given in the title. Bottom: the rear axle's lateral error along the course. Last frame: within 0.1 m after 4.1 m, 0.04 m past the line, 0.000 m in the middle of the arc

5.3 Why it converges: a spring and a damper over distance

Near a straight course the law is pure pursuit's spring and damper again, with its two gains chosen directly. As there, we follow the error along the distance driven. With the lateral

error e, the heading error θ_e and the gains k_θ and k_e of Section 5.2, for small angles and forward driving, and:

- e', how fast the lateral error changes per metre driven: the heading error θ_e;
- e'', how fast e' changes per metre: the turn rate over the speed.

Dividing the turn rate by the speed, on a straight course:

$$e'' = -1.0\, e' - 0.5\, e$$

$$e'' + 1.0\, e' + 0.5\, e = 0$$

$$e'' + k_\theta\, e' + k_e\, e = 0$$

Matched to the natural frequency and damping ratio (G-2991, G-2992), per metre driven:

$$\omega_n^2 = 0.5$$

$$\omega_n = 0.7071 \text{ per m}$$

$$\zeta = 1.0/(2 \times 0.7071)$$

$$= 0.707$$

$$\omega_n = \sqrt{k_e}$$

$$\zeta = \frac{k_\theta}{2\sqrt{k_e}}$$

With a damping ratio of 0.707 the overshoot is 4.3 percent, 0.043 m from a 1.0 m start; the run crosses by 0.039 m (notebook). The speed has divided out, so the car returns over the same distance at any speed: within 0.1 m after 4.1 m at 2 m/s (2.1 s) and 4.0 m at 5 m/s (0.83 s). Paden et al. multiply the heading gain by the size of the speed for exactly this reason (§V.A.2).

On the arc the curvature part turns the car with the course, so the errors stay at zero: 0.000 m in the middle of the arc and 0.005 m after it (notebook).

Backwards, the speed is negative. The heading part keeps its sign, because it uses the size of the speed, so it still damps. The lateral part changes sign with the speed, and so does the

way the lateral error changes as the car rolls, so the part still pushes the error towards zero. Paden et al. (§V.A.2) state that the law converges for either direction of travel, shown with an energy-like function of the two errors that can only fall; such functions are the tool of Kanayama's tracker. In Figure 7 the reversing car returns over 4.1 m with 0.039 m past the line, the same as forwards.

6. Where they run

> **Key point:** Stanley steered Stanford's car to win the 2005 DARPA Grand Challenge, at 20 commands a second with a softening speed of 1 m/s. Autoware and Nav2 ship neither Stanley nor rear-wheel feedback: Autoware's lateral controllers are model predictive control and pure pursuit, Nav2's are regulated pure pursuit, MPPI and others.

Stanley in the desert. Stanley "won the 2005 DARPA Grand Challenge", a race for driverless cars across the desert (Thrun et al. 2006). Its steering controller took the planner's trajectory, the estimated pose and speed and the measured steering angle, and sent steering commands 20 times a second (Thrun et al. §9.2), the rate our trackers use. The law was Section 2's with a softening speed of 1 m/s and the damping terms of Section 4.3; Stanford's team later drove its Urban Challenge car, Junior, with it at up to 71 mph on pavement (Hoffmann et al. §III.A). MATLAB's Automated Driving Toolbox ships a Stanley lateral controller block, in kinematic and dynamic versions, with a position gain of 2.5 by default and a steering limit of 35° (MathWorks Lateral Controller Stanley documentation).

Not in Autoware, Nav2 or Apollo. None of the three open stacks ships either tracker:

- **Autoware:** its trajectory follower picks its lateral controller with `lateral_controller_mode`, which is either `mpc` (model predictive control) or `pure_pursuit` (`controller_node.cpp` in `autoware_trajectory_follower_node`);
- **Nav2:** its controller plugins are regulated pure pursuit, MPPI (its default in `nav2_params.yaml`), DWB, the graceful controller and the rotation shim;
- **Apollo:** it steers with an LQR controller (`lat_based_lqr_controller`) or MPC.

Open code. PythonRobotics implements both: `stanley_control.py`, with gain 0.5 per s, the error at the front axle and no softening speed; and `rear_wheel_feedback_control.py`, which is Paden's eq. V.12 with the gains 1.0 and 0.5 used here.

7. Summary

Tracker	Measures at	Steering or turn rate	Our numbers
Pure pursuit (RO-071)	rear axle	arc to the point 4 m ahead	0.12 m inside the arc, 0.17 m wide after it
Stanley	front axle	$\delta = -\theta_e - \tan^{-1}(ke/(k_s + v))$	error halves in 1.03 s at 2 m/s, 0.81 s at 5 m/s; 0.004 m mid-arc
Rear-wheel feedback	rear axle	curvature part + heading part + lateral part, then $\delta = \tan^{-1}(L\omega/v)$	within 0.1 m after 4.1 m at any speed; 0.000 m mid-arc; works backwards

- The opening problem has its answer: pure pursuit's rear axle ran 0.12 m inside the arc and 0.17 m wide after it, while Stanley's front axle is 0.004 m off in the middle of the arc and rear-wheel feedback's rear axle 0.000 m, because both use the course's direction at the car rather than at a point ahead.
- Stanley points the front wheels parallel to the course and then towards it, so they aim at the course point a set distance ahead, the speed plus the softening speed over the gain: 3 m at 2 m/s, because that one angle shrinks the front axle's error directly.
- The front wheel rolls the way it points, so Stanley's front-axle error shrinks in proportion to itself, a first-order system whose rate is the gain times the speed over the speed plus the softening speed (0.667 per s at 2 m/s): on a car whose wheels turn at once it never swings, and it takes a similar time at any speed, so the fast car uses more road (4.0 m against 2.0 m to halve the error).
- On the arc Stanley's heading term supplies the 14.5° the curve needs, so its front axle stays on the course while its rear axle runs 0.32 m inside; trackers must be compared at the same point of the car.
- A larger gain returns faster, but with a steering motor that lags 0.4 s, which we added to the chapter's car because real steering lags, gains of 2.5 and 5 per s swing past the line by 0.24 and 0.46 m, because the law acts on errors the lagging wheels answer late, so the gain has an upper limit.
- The softening speed caps the gain on the lateral error at the gain over the softening speed, 1 per m, so a 0.05 m jump in the measured error turns the wheels 1.9° instead of 5.7° at 0.5 m/s.
- Rear-wheel feedback adds the turn the path's curvature needs to feedback on the heading and lateral errors, so it holds the arc with no error and, near the line, is a spring and a damper over distance with a damping ratio of 0.707, settling within 4.1 m at any speed.
- Rear-wheel feedback works backwards and Stanley does not, because the rear axle's errors keep shrinking for either sign of the speed while the trailing front axle turns the car round.

The car can now follow a path through its corners, forwards or backwards, with no look-ahead to retune. Still missing: every tracker so far follows a path with no clock, and a plan that

says where the car must be at each moment needs a tracker that also keeps to time, the tracking of a timed trajectory.

8. Sources

Built from

- MATLAB, "Vehicle Path Tracking Using Stanley Controller", YouTube, https://www.youtube.com/wat JBs
- Snider, J. M. (2009). *Automatic Steering Methods for Autonomous Automobile Path Tracking*. Technical report CMU-RI-TR-09-08, Robotics Institute, Carnegie Mellon University. §1 Introduction (tests on the CarSim simulator with steering delays), §2.3 Stanley method (eq. 5, Fig. 14), §2.3.1 Tuning the Stanley controller. https://www.ri.cmu.edu/pub_files/2009/2/Automatic_Steering_Methods_for_Autonomous_Autom (Snider)
- Paden, B., Čáp, M., Yong, S. Z., Yershov, D. and Frazzoli, E. (2016). "A Survey of Motion Planning and Control Techniques for Self-driving Urban Vehicles". *IEEE Transactions on Intelligent Vehicles* 1(1). §V.A.2 Rear wheel position based feedback (eqs. V.10–V.12), §V.A.3 Front wheel position based feedback (eqs. V.13–V.15). https://arxiv.org/abs/1604.07446 (Paden)
- Hoffmann, G. M., Tomlin, C. J., Montemerlo, M. and Thrun, S. (2007). "Autonomous Automobile Trajectory Tracking for Off-Road Driving: Controller Design, Experimental Validation and Racing". *American Control Conference*. §III.A Lateral control (eqs. 5, 7, 9). http://ai.stanford.edu/~gabeh/papers/hoffmann_stanley_control07.pdf (Hoffmann et al.)
- Thrun, S. et al. (2006). "Stanley: The Robot that Won the DARPA Grand Challenge". *Journal of Field Robotics* 23(9), 661–692. §9.2 Steering control (eqs. 7–9). http://robots.stanford.edu/papers/thrun.stanley05.pdf (Thrun et al.)
- Code read for Sections 5.2 and 6:
  - PythonRobotics: PathTracking/rear_wheel_feedback_control/rear_wheel_feedback_control.py in AtsushiSakai/PythonRobotics
  - PythonRobotics: PathTracking/stanley_control/stanley_control.py in AtsushiSakai/PythonRobotics
  - Autoware: control/autoware_trajectory_follower_node/src/controller_node.cpp in autowarefoundation/autoware_universe
  - Nav2: nav2_bringup/params/nav2_params.yaml in ros-navigation/navigation2
  - Apollo: modules/control/controllers/lat_based_lqr_controller/lat_controller.cc in ApolloAuto/apollo

Other references

- MathWorks, "Lateral Controller Stanley" block documentation, Automated Driving Toolbox (vehicle models, gains, maximum steering angle), https://www.mathworks.com/help/driving/r (MathWorks Lateral Controller Stanley documentation)

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Stanley controller (front-wheel feedback) (G-3160)	A path tracker that measures the lateral and heading errors at the front axle and steers the front wheels parallel to the path and then towards it by $\tan^{-1}(ke/(k_s + v))$, aiming them at a path point ahead; the front-axle error then shrinks by the same fraction each second, but the law is not stable when reversing.
Softening speed (Stanley) (G-3161)	A constant k_s, such as 1 m/s, added to the speed in the Stanley controller's cross-track term, so its gain on the lateral error never exceeds k/k_s at low speed; it stops noise in the measured error from shaking the wheels when the car is slow.
Rear-wheel feedback (G-3162)	A path tracker that measures the errors at the rear axle and asks for a turn rate made of the path's own turn (the speed times the path curvature) plus parts against the heading error and the lateral error, turned into steering by the bicycle model; it holds curves with no steady error and still works when the car reverses.
Pure pursuit (G-3150)	A path tracker that, at every tick, finds the point on the path one look-ahead distance from the rear axle and steers along the circle that leaves the car along its heading and passes through that point, of curvature $2y/l_d^2$; looking ahead brings the heading error into the steering, so a car whose steering answers at once returns to the path with little swing past it.
Look-ahead point (pure pursuit) (G-3152)	The point of the path one look-ahead distance from the rear axle, found by walking forward from the nearest path point, such as (3.873, 0) for a car 1 m off a straight line with a 4 m look-ahead; pure pursuit steers along the arc that reaches it. Also called the goal point or carrot.

Term	Meaning
Look-ahead distance (pure pursuit) (G-3151)	The distance l_d from the rear axle to the point pure pursuit chases, such as 4 m; a short one brings the car back sooner but weaves when the steering lags, a long one is smooth but cuts corners.
Wheelbase L (G-3127)	The distance from a car's rear axle to its front axle, such as 2.5 m; with the steering angle it fixes the turning radius, $R = L/\tan\delta$, so a longer car turns more widely for the same steering.
Kinematic bicycle model (G-3128)	A car drawn as one steered front wheel and one rear wheel on its centre line, with no tyre slip: the rear axle moves along the heading and the heading turns at $\dot{\theta} = v\tan\delta/L$; it predicts a car's motion well while the sideways acceleration stays below about half the road's friction (grip) coefficient times gravity.
Lateral error e (cross-track error) (G-3135)	The signed distance from a path to a point of the vehicle, measured along the path's normal at the nearest path point and positive to the left, such as -0.770 m for a car outside an arc; it is one of the two numbers a path tracker drives to zero.
Heading error θ_e (G-3136)	A vehicle's heading minus the path's direction at the nearest path point, wrapped into $-180°$ to $180°$, such as $10° - 21.8° = -11.8°$; it says whether the vehicle is turning back to the path or away, and the lateral error changes at $v\sin\theta_e$.
First-order system (G-2987)	A system whose output follows a differential equation with only a first derivative, such as the robot's speed under P control; it closes the same fraction of its remaining gap in every time constant and can never overshoot.
Time constant (G-2988)	The time a first-order system takes to close 63 % of the gap to its final value, m/(K_v + b) = 1 s for the robot's speed loop; it fixes the rise time (about 2.2 time constants) and the 2 % settling time (about 4).

Term	Meaning
Path curvature κ (G-3129)	How fast a path's direction turns per metre along it, $1/R$ on an arc of radius R, such as 0.1 per m on a 10 m arc and 0 on a straight; it fixes the steering a bend needs, $\tan\delta = L\kappa$.
Feedforward (G-3070)	A control force computed from the plan and the robot's model, such as mass times the planned acceleration plus drag ($ma_r + bv_r$), and sent before any error appears; it lets the robot follow a changing plan without first falling behind, while feedback removes what the model gets wrong.
Natural frequency (G-2991)	The rate, in rad/s, at which a second-order system would swing with no damping, $\omega_n = \sqrt{K_p/I}$, 2 rad/s for the heading loop; it sets how fast the reply is, and raising it squeezes the whole reply in time.
Damping ratio (G-2992)	A second-order system's damping compared with the damping that just stops it from swinging, $\zeta = K_d/(2\sqrt{K_pI})$, 0.2 for D gain 0.08 and 1 for 0.4; it alone sets the shape of the reply and its overshoot.